1.Branch Instruction Verification

```
===== BRANCH INSTRUCTION CHECKS =====
PASS: x3 = 1
PASS: x4 = 2
PASS: x5 = 3
PASS: x6 = 4
PASS: x7 = 5
PASS: x8 = 6
PASS: x9 = 7
PASS: x10 = 8
PASS: x11 = 9
PASS: x12 = 10
PASS: x13 = 11
PASS: x14 = 12

===== LOAD-USE BRANCH HAZARD =====
PASS: x16 = 5
PASS: x17 = 13

===== JALR ALIGNMENT =====
PASS: x19 = 136
PASS: x20 = 15
```

```
===== LOAD-USE BRANCH HAZARD =====
PASS: x16 = 5
PASS: x17 = 13

===== JALR ALIGNMENT =====
PASS: x19 = 136
PASS: x20 = 15

===== INITIAL TEST SUMMARY =====
PASS: ALL EDGE CASE CHECKS

===== RESET CHECK =====
PASS: x1 = 0
PASS: x3 = 0
PASS: x16 = 0
PASS: x19 = 0
PASS: x1 = 5
PASS: x3 = 1
PASS: x16 = 5

==================================
 PASS: ALL EDGE CASE TESTS
==================================
```

>>CORRESPONDING INSTRUCTION CODE:

```
00500093
00500113
FFF00793
09100913
00102023
00002803
00180263
00D00893
00208263
00F08263
00F09263
00209263
0017C263
00F0C263
00F0D263
0017D263
00F0E263
0017E263
0017F263
00F0F263
00100193
00200213
00300293
00400313
00500393
00600413
00700493
00800513
00900593
00A00613
00B00693
00C00713
00000013
000909E7
00000013
00000013
00F00A13
```

2. Branch Hazard, JALR and Reset Verification

```
================================================
   5-STAGE PIPELINE BRANCH VERIFICATION
================================================


------ DEBUG ADDI x15, x0, -1 ------
opcode_ex          = 0010011
rd_ex              = 15
rs1_ex             =  0
forwarded_rs1       = 00000000
imm_ex             = ffffffff
alu_src_ex         = 1
alu_operand_a      = 00000000
alu_operand_b      = ffffffff
alu_control_signal = 0000
alu_result_ex      = ffffffff
-----------------------------------
```

```
WB DEBUG: x15 <= ffffffff
BRANCH 0 | funct3=000 | taken=1 | PC=00000018
PASS: Branch 0
BRANCH 1 | funct3=000 | taken=1 | PC=00000020
PASS: Branch 1
BRANCH 2 | funct3=000 | taken=0 | PC=00000024
PASS: Branch 2
BRANCH 3 | funct3=001 | taken=1 | PC=00000028
PASS: Branch 3
BRANCH 4 | funct3=001 | taken=0 | PC=0000002c
PASS: Branch 4
BRANCH 5 | funct3=100 | taken=1 | PC=00000030
PASS: Branch 5
BRANCH 6 | funct3=100 | taken=0 | PC=00000034
PASS: Branch 6
BRANCH 7 | funct3=101 | taken=1 | PC=00000038
PASS: Branch 7
BRANCH 8 | funct3=101 | taken=0 | PC=0000003c
PASS: Branch 8
BRANCH 9 | funct3=110 | taken=1 | PC=00000040
PASS: Branch 9
BRANCH 10 | funct3=110 | taken=0 | PC=00000044
PASS: Branch 10
BRANCH 11 | funct3=111 | taken=1 | PC=00000048
PASS: Branch 11
BRANCH 12 | funct3=111 | taken=0 | PC=0000004c
PASS: Branch 12
```

>>CORRESPONDING INSTRUCTION CODE:

00500093
00500113
FFF00793
09100913
00102023
00002803
00180263
00D00893
00208263
00F08263
00F09263
00209263
0017C263
00F0C263
00F0D263
0017D263
00F0E263
0017E263
0017F263
00F0F263
00100193
00200213
00300293
00400313
00500393
00600413
00700493
00800513
00900593
00A00613
00B00693
00C00713
00000013
000909E7
00000013
00000013
00F00A13

3.Load/Store and Load-Use Hazard Verification

02A00093
00102023
00002103
00110193
00000013
00000013
00000013
00000013
00000013
00000013

```
============================================
 LOAD / STORE + LOAD-USE HAZARD VERIFICATION
============================================
STORE DEBUG: addr=00000000 data=0000002a funct3=010
LOAD DEBUG: addr=00000000 read_data=0000002a funct3=010

------ REGISTER CHECKS ------
PASS: x1 = 42
PASS: x2 = 42 (LW)
PASS: x3 = 43 (load-use hazard)

------ MEMORY CHECKS ------
PASS: memory byte 0 = 2A
PASS: memory byte 1 = 00
PASS: memory byte 2 = 00
PASS: memory byte 3 = 00

============================================
PASS: ALL LOAD/STORE TESTS PASSED
============================================
```

4.ALU Data Forwarding Verification

00A00093
00508113
00310193
00700213
00900293
00520333
001303B3
00000013
00000013
00000013
00000013
00000013
00000013
00000013
00000013
00000013

```
============================================
 ALU DATA FORWARDING VERIFICATION
============================================

------ REGISTER CHECKS ------
PASS: x1 = 0000000a
PASS: x2 = 0000000f
PASS: x3 = 00000012
PASS: x4 = 00000007
PASS: x5 = 00000009
PASS: x6 = 00000010
PASS: x7 = 0000001a

============================================
PASS: ALL ALU FORWARDING TESTS PASSED
============================================
```

5.Load/Store Edge Case Verification

```
================================================
 LOAD / STORE EDGE CASE VERIFICATION
================================================

------ REGISTER CHECKS ------
PASS: x1 = 00000080
PASS: x2 = 00001234
PASS: x3 = 89abcdef
PASS: x4 = ffffff80
PASS: x5 = 00000080
PASS: x6 = 00001234
PASS: x7 = 00001234
PASS: x8 = 89abcdef
PASS: x9 = 89abcdf0

------ MEMORY BYTE CHECKS ------
PASS: mem[0] = 80
PASS: mem[1] = 00
PASS: mem[2] = 34
PASS: mem[3] = 12
PASS: mem[4] = ef
PASS: mem[5] = cd
PASS: mem[6] = ab
PASS: mem[7] = 89

================================================
PASS: ALL LOAD/STORE EDGE CASES PASSED
================================================
```

```
08000093
00001137
23410113
89ABD1B7
DEF18193
00100023
00201123
00000013
00000013
00000013
00302223
00000203
00004283
00201303
00205383
00402403
00140493
00000013
00000013
00000013
00000013
00000013
```

6.Pipeline Integration Verification

```
============================================
 COMBINED PIPELINE VERIFICATION
============================================

------ REGISTER CHECKS ------
PASS: x1 = 0000002a
PASS: x2 = 0000002a
PASS: x3 = 00000034
PASS: x5 = 0000000a
PASS: x6 = 00000005
PASS: x7 = 00000008
PASS: x8 = 0000000d
PASS: x9 = 00000005
PASS: x10 = 00000005
PASS: x11 = 00000005
PASS: x12 = 00000000
PASS: x13 = 0000000a

------ MEMORY CHECKS ------
PASS: mem[0] = 2a
PASS: mem[1] = 00
PASS: mem[2] = 00
PASS: mem[3] = 00
PASS: mem[4] = 2a
PASS: mem[5] = 00
PASS: mem[6] = 00
PASS: mem[7] = 00

============================================
PASS: ALL COMBINED PIPELINE TESTS PASSED
============================================
```

00002083
00102223
00402103
00A00293
005101B3
00500313
00330393
00638433
407404B3
00500513
00500593
00B50463
06300613
00A00693
00000013
00000013
00000013
00000013
00000013
00000013

7.Reset and x0 Register Verification

```
============================================
 RESET / X0 VERIFICATION
============================================
PASS: x0 = 00000000
PASS: x1 = 00000007
PASS: x2 = 00000009
PASS: x3 = 0000000b
PASS: x4 = 0000000d
PASS: x5 = 0000000f
PASS: x6 = 00000011
PASS: x7 = 00000013
PASS: x8 = 00000015
PASS: x9 = 00000017

============================================
PASS: RESET / X0 CHECKS PASSED
============================================
```

00500013
00700093
00900113
00B00193
00D00213
00F00293
01100313
01300393
01500413
01700493